

Series and Parallel Circuits, Ohm’s Law and Non-Ohmic Resistors

DC Circuits: Series, Parallel & Ohm’s Law — Custom (Claude AI)

Year 11	~30 minutes	Science · Physics
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NOTICE — AI-generated worksheet

This worksheet was drafted automatically from the site’s curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: [simulations/dc-circuits-and-ohms-law.html](#) — open it from the STEM Simulation Hub

Curriculum links: QCAA Physics — Unit 1, Topic 3: Electrical circuits (electric charge and current as the flow of charge carriers, conservation of charge · potential difference and Ohm’s Law · resistance, and ohmic vs non-ohmic behaviour · series and parallel circuit rules for current, voltage and resistance · circuit diagrams, voltmeters and ammeters · electrical energy and power)

Learning intentions

- I can calculate the total resistance of resistors in series, in parallel, and in simple series–parallel combinations.
- I can apply Ohm’s Law ($V = IR$) to find an unknown voltage, current or resistance in a circuit.
- I can use a virtual voltmeter and ammeter correctly, and explain why a non-ohmic component such as a lamp or diode doesn’t have a constant resistance.

Getting started

Open the DC Circuits simulation and stay on the Circuit Builder tab. Choose a topology from the Topology panel on the left, then set each resistor’s value either by clicking it directly on the diagram or from the side panel — any resistor can also be swapped for a non-ohmic lamp or diode. Drag the source voltage slider to change the battery (or click the battery on the diagram), and use the Meter toggle to switch on the voltmeter or ammeter and click points on the diagram to place its probes. The Current–voltage characteristics graph below the circuit plots every component’s I–V curve live, with its exact operating point marked. This worksheet uses Circuit Builder for Tasks 1–6, and Quiz Mode for Task 7.

Tasks

1. Select the **Series ×2** topology. Set both resistors to ohmic, with $A = 5\ \Omega$ and $B = 10\ \Omega$, and set the source voltage to 15 V. By hand, use $V = IR$ to calculate the total (equivalent) resistance and the current supplied by the source. Check both against the Equivalent R and Total current readouts. ^[3]

Working:

2. Staying on this circuit, switch the meter to Voltmeter and click the two ends of resistor B to read the voltage across it. Using the current you calculated in Task 1, calculate by hand what this voltage reading should be, then compare it to what the simulation shows. [3]

Working:

3. Switch to **Parallel ×2** with A = 6 Ω and B = 12 Ω, and source voltage 12 V. By hand, calculate the equivalent resistance using $1/R_{\text{total}} = 1/R_A + 1/R_B$, then compare it with the Equivalent R readout. Switch to Ammeter and click a point in each branch — record both currents and show that they add up to the Total current. [4]

Working:

4. Using the diagram and meters from Task 3, complete this table for the Parallel ×2 circuit:

Resistor	Resistance (Ω)	Voltage across it (V)	Current through it (A)
A	6		
B	12		

What do you notice about the voltage across A compared to B? Explain why, in terms of what the two ends of a parallel branch always share.

[3]

5. Change either resistor slot to Lamp and look at the Current–voltage characteristics graph. Describe how the lamp’s curve differs in shape from an ohmic resistor’s straight line, and explain what this tells you about how the lamp’s resistance changes as the current through it increases. [2]

6. Choose one of the combined topologies (A + (B || C) or (A || B) + C) and set all three resistors to ohmic values of your choice. By hand, calculate the equivalent resistance step by step — the parallel part first, then add the series part — showing every step, then check your answer against the simulation. [4]

Working:

7. Switch to **Quiz Mode** and set Difficulty to Combined. Answer three questions in a row, writing down your working for each *before* pressing Check. Record all three questions, your working, and whether each answer was correct. [4]

Working:

Extension (optional)

A student wires a diode in series with a resistor and a 6 V source, then keeps increasing the resistor's value. Using the diode's current–voltage curve on the graph — near-zero current below its turn-on voltage, then a steep, non-linear rise above it — predict qualitatively what happens to the current in the circuit as the series resistance keeps increasing. Explain why a series resistor is often needed to protect a real diode or LED in a circuit.